

Ojas Jeetendra Raundale

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EDUCATION

- University of Massachusetts Amherst** Massachusetts, US
Masters of Science (MS) - Computer Science; GPA: 3.87/4.0 Sept 2023 - May 2025 (exp.)
Courses: Reinforcement Learning, Natural Language Processing, Probabilistic Graphical Models, Big Data, Distributed Systems
- Indian Institute of Technology Dharwad** Karnataka, India
Bachelor of Technology - Computer Science and Engineering; July 2017 - June 2021

EXPERIENCE

- Fidelity Investments** Boston, MA
Data Scientist Jun 2025 - Present
 - Developed a natural-language-to-structured-query generation pipeline for the "Chat with my Data" flow by **fine-tuning LLMs (SFT, RLHF)** to generate schema-aware JQ filters, enabling targeted extraction of relevant subsets from large customer datasets.
 - Designed a **repeatability-focused evaluation** framework using entropy-based and population-level distributional metrics to **quantify non-determinism** and drive improvements in output consistency for LLM-based systems.
 - Improved LLM **tool-calling** for **agentic workflows** by designing evaluation benchmarks and grounded **synthetic datasets** to compare generated tool calls against gold standards; evaluated models across accuracy, latency, and throughput, and refined tool definitions and prompts for reliable production behavior.
- Google DeepMind — IESL Lab, University of Massachusetts, Amherst** Remote, USA
Graduate Research Extern Jan 2025 - May 2025
 - Researched and developed **Quadratic Model Ensembling** methods for LLMs and VLMs using adaptive optimization techniques in collaboration with **Google DeepMind**
 - Investigating Out-of-Distribution performance, and computational expenditure of Ensembling Methods in comparison with fine-tuning and Model SOUPing.
- IBM — IESL Lab, University of Massachusetts, Amherst** Amherst, MA
Graduate Research Extern August 2024 - May 2025
 - Developing and researching **fine-tuning algorithms for LLMs** based on distribution matching methods under Prof. Andrew McCallum and in collaboration with **IBM Research**.
 - Utilised concepts from **Markov Chains** and **Self-Play Optimizations** to achieve accuracies better than Direct Preference Optimization (DPO) and Reinforcement learning from human feedback (RLHF).
- FCAT, Fidelity Investments** Boston, MA
Data Science Intern June 2024 - August 2024
 - Reduced ChatGPT API costs by more than 95% by replacing it with in-house **fine-tuned Large Language Models**.
 - Automated the process of support-customer phone call evaluation by fine-tuning **Phi-3** and **Llama 3.1** using **LORA** and **qLORA** and achieving accuracies greater than 90%.
 - Designed custom LLM evaluation metrics and utilized a multi-GPU distributed system to train LLMs using **Deepspeed**.
- Express Analytics** Pune, India
Machine Learning Engineer - Data Science June 2021 - June 2023
 - Improved customer targeting and marketing effectiveness resulting in increased ROI by developing custom **Multi-Touch Attribution and Marketing Mix Models**.
 - Analyzed millions of customer journeys from **Google Analytics Data** by employing models built upon RNNs, LSTMs, and Statistical Models in **Python's Keras** to optimize marketing campaign spends.
 - Revamped the deployment procedures by creating **CI/CD pipelines** and **Dockerized microservices** on **Jenkins** DevOps and **Gitlab** reducing deployment effort from days to hours.

PROJECTS

- Medical Minds Unite - Under guidance of Prof. Mohit Iyyer (NLP, Transformers, PyTorch) ** :
Strengthened the performance of open-source LLMs for medical applications by fine-tuning, quantizing, and implementing merging techniques such as Ensembling, Mixture of Experts, and QLoRa.
- Statistical Models Repository (JAX, PyTorch, Monte Carlo Methods): **
Developed a repository of statistical model implementations to explore innovative methods for solving complex problems such as Generation and Bradley-Terry Ranking. Experimented with Hamiltonian MCMC, Message Passing, Conditional Random Fields, and Variational Autoencoders resulting in robust and effective solutions.
- Reinforcement Learning Repository (Deep Learning, Reinforcement Learning, PyGame, PyTorch): **
Assembled a Portfolio of Reinforcement Learning Algorithm implementations like Policy Gradient methods, Q-learning, SARSA, and Evolution strategies. Trained and evaluated RL agents on classical RL domains such as cart pole and lunar lander as well as self-made games helping the AI bot play the game with optimal performance.

PUBLICATIONS

- **Earnings2Insights: Persuasive Investment Report Generation Using Single And Multi-Agent Frameworks (FinNLP @ EMNLP 2025) **:
Co-authored a paper on LLM-based multi-agent frameworks for generating investment reports; ranked 2nd overall and 1st in readability in the Earnings2Insights task at FinNLP.

SKILLS SUMMARY

- **Languages:** Python, C/C++, Java, SQL, R, Bash
- **ML / GenAI:** LLMs, Fine-tuning, LLM Evaluations, Synthetic Datasets, Agentic AI, Distributed Training
- **Frameworks:** PyTorch, JAX, TensorFlow, HuggingFace Transformers, Keras, PySpark, vLLM
- **Systems & Tools:** Docker, AWS, DeepSpeed, Git, Power BI, Tableau